the sulphur at the point of contact with the phosphorus sustained a considerable loss of substance. Similar conditions being found at the end of three months, the phosphorus still fluid, and the sulphur having a continued waste, the amount of electricity generated was tested by Thomson's electrometer, and the electric motive force was 162°, in comparison with a Daniell cell, which only gave 120°; the difference was 42° in favor of the new cell. The constancy of this original battery was shown by its continuing to work steadily after the expiration of several months.

Mr. Gray proceeds from these hints to suggest a new hypothesis in regard to nerve-force; and starting with the assumption that nerve-power has in it an electric element, he endeavored to ascertain its source, which, after elaborate experiment, he believes can be found in the sulphur and phosphorus of the human body: the brain contains considerable phosphorus, whilst sulphur exists in the liver, and an alkaline solution is circulated between them. He took a frog, and having secured anæsthesia by application of chloroform, an incision was made through the abdominal walls in the right hypochondriac region; and a copper wire passed into the substance of the liver. The eyeball was then pierced, and a similar piece of copper wire brought in contact with the brain by passing it through the optic foramen. The free extremities of the copper wires were brought in contact with the exposed sciatic nerve of another frog's hind leg, when powerful convulsions were immediately induced in the muscles.